CT117-3-2-FWDD Further Web Design and Development

Individual Assignment

Assignment Objectives

- To design and implement a multi-page web interface using modern UI/UX principles.
- To develop a web app using Node.js, complete with a well-defined API for client-server communication.
- To integrate a SQL-based data persistence system to store and retrieve application data, ensuring seamless communication between the server and the database.
- To incorporate standard security mechanisms to authenticate users, safeguarding user data and enhancing the overall security of the web app.

Assignment Brief

As an aspiring Software Engineer, your task is to create a **prototype of an educational hybrid game** aimed at **assessing a specific Python programming concept** through **formative assessment**. An educational hybrid game combines elements of traditional learning methods with game-based learning techniques. This approach **integrates physical and digital components** to create an engaging and interactive learning experience. The game should offer engaging formative assessment activities such as Interactive Quizzes, Skill Assessments, or Knowledge Tests to assess and enhance users' proficiency in the chosen Python concept. The educational hybrid game should support **multiplayer** functionality, allowing users to compete or collaborate in real-time.

In this assignment, the **web app will serve as the digital component** of the educational hybrid game. It could be used to display interactive quizzes, skill assessments, or knowledge tests related to the chosen Python concept. The web app could also serve as a scoreboard, tracking and displaying players' scores in real time.

The **physical component, like a board game or card game**, complements the digital component. There must be a **mechanism such as QR Code, RFID, NFC, or any other suitable technology** to connect the physical and digital components, ensuring interaction between them.

You may choose to adapt an existing game or create a new game.

Adapting Existing Games

You may choose to adapt an existing game (e.g., Snakes and Ladders, Chess, Monopoly) with a twist of digital component (the web app). For example, each square on the Snakes and Ladders board could contain a QR code. Players would need to scan the QR code using a web app to reveal a Python programming concept-related question. If they answer the question correctly, they could be rewarded with a boost (like climbing a ladder). If they answer incorrectly, they might face a setback (like sliding down a snake). The digital component could provide immediate feedback on the player's answers and update the game state accordingly. The rules of Snakes and Ladders would remain the same, but the addition of Python-related questions and the use of a web app would add an educational twist to the game.

Creating New Educational Hybrid Games

Alternatively, you can create your own educational hybrid game with your own rules, as long as it fulfils the requirements of this assignment. The specific scope and content of the game are left to your creative discretion.

Requirements

The educational hybrid game must fulfill the following criteria:

i) Database interaction is a must. This includes capabilities to add, retrieve, update, and delete data as required.

ii) User recognition should be incorporated. This implies the necessity for user login and data preservation for future visits.

iii) It should be engaging and user-friendly, responding promptly to user actions.

iv) Form input validation is essential to maintain the integrity and reliability of your data.

v) Responsive web design is crucial to ensure a consistent experience across all devices, be it a phone, tablet, or computer.

Assignment Stages

Stage	Activity	Milestone	Submission
Analysis	1. Choose a Python Concept ▪ Select a specific Python programming concept that the game will focus on. This could be anything from basic syntax and data types to more advanced topics like object-oriented programming or data structures. 2. Identify the role of the Web App as Digital Component ▪ Determine how the web app will be used in the game. This could be to display questions, track and display scores, provide feedback, etc. 3. Identify the Game Physical Component ▪ Decide on the physical component of the game. It could be a physical game board, cards etc.	▪ Project Proposal i) Title ii) The chosen Python programming concept (please choose 1 concept only) iii) The role of the Web App as the Digital Component of the educational hybrid game iv) The physical components of the game. v) Specifications for the end user	15th March 2026
Design	Design the Storyboard and Database Diagram (ERD and Data Dictionary) ▪ Sketch out a storyboard for the web app to plan its layout and user interface to visualize the sequence of events in the game and how the user will navigate	▪ Preliminary Draft	19th April 2026

	through it. Also, design a database diagram to outline how game data will be stored and managed.		
Development	▪ Develop the educational hybrid game based on the proposal, storyboard and database design. ▪ Prepare the final report according to the suggested contents in the section.	▪ Educational Hybrid Game Web App ▪ Database file (i.e. in SQL file format) ▪ Final Report	31^{st} May 2026

Final Report

- Format:
  - Typeface
    - Please use Times New Roman. You may use bold, italic, and underlined text for emphasis and to enhance readability.
  - Font size
    - The standard font size should be 12, except for titles and headings.
  - Spacing
    - Maintain a spacing of 1.5 lines between the texts within a paragraph.
  - Alignment
    - The text should be justified.
  - Headers and footers can be utilized for additional information.
  - The document should be numbered for easy reference.
  - A cover page is required. Your cover page should include the following details:
    - Your Name and ID
    - Intake code
    - Subject
    - Project Title

- Recommended Report Content
  - **Table of Contents**
    - The table of contents should list each topic title along with its corresponding page number.
  - **Introduction/Project Plan**
    - This section should introduce your project, including its objectives, scope, end-user specifications, and the major functions of the web app.
  - **Design**
    - Please include an Entity Relationship Diagram (ERD), Data Dictionary, and Storyboard for the main features of the educational hybrid game. Each component should be accompanied by an appropriate description.
  - **Implementation**
    - This section should provide a detailed explanation of the source codes for the major features of the web app.
  - **User Guidance**
    - Step by step on how to play the hybrid game.
    - Include screenshots of the user interface (physical and digital components of the game) and descriptions of each.
  - **Conclusions**
    - Summarize your project and discuss potential future enhancements.
  - **References**
    - List all sources referenced in APA format.

Final Submission

- The complete project and report to be submitted via Moodle.

Presentation

- A presentation is scheduled to take place following the submission (Week of 1st June 2026). The allocation of presentation slots will be announced as we approach the end of the semester.